

2 The photograph shows part of a desert in Mexico.



(Source: © Alexander Semkin/Alamy Stock Photo)

A scientist observed that one species of cactus, *Neobuxbaumia tetetzo*, was often found near a species of tree, *Mimosa luisana*.

The scientist decided to test the hypothesis that the cactus plants often grow close to mimosa trees.

One hundred locations were selected at random. A 10×10 m quadrat was placed at each location.

The presence or absence of each species was recorded in each quadrat.

(a) Suggest one possible benefit to cactus plants of growing near mimosa trees.

(1)

(b) Suggest **one** risk the scientist might encounter when carrying out this investigation and how you could reduce this risk.

(2)

Risk

How to reduce the risk

(c) The table shows the results of this investigation.

Mimosa trees	Number of quadrats	
	Cactus plants present	Cactus plants absent
Present in quadrat	60	4
Absent in quadrat	16	20

- (i) An odds ratio can be used to determine if the presence of mimosa trees has an effect on the presence of the cactus plants.

Calculate an odds ratio using the following steps.

Give your answer to **two** significant figures.

$$\text{Step 1} = \frac{\text{Number of quadrats in which mimosa trees and cactus plants are present}}{\text{Number of quadrats in which mimosa trees are absent and cactus plants are present}}$$

$$\text{Step 2} = \frac{\text{Number of quadrats in which mimosa trees are present and cactus plants are absent}}{\text{Number of quadrats in which both mimosa trees and cactus plants are absent}}$$

$$\text{Odds ratio} = \text{Step 1} \div \text{Step 2}$$

(3)

(ii) The table shows how the odds ratio can be used in this investigation.

Odds ratio	Presence of mimosa trees
< 1	Reduces the likelihood of cactus plants being present
$= 1$	Has no effect on the likelihood of cactus plants
> 1	Increases the likelihood of cactus plants being present

Give a conclusion that can be made from the results of this investigation.

(1)

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(d) Abiotic variables in the soil cannot be controlled in this investigation.

However, these variables can be measured to confirm that these plants are growing in similar conditions.

(i) State **two** abiotic variables in the soil that could affect this investigation.

(2)

First variable

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Second variable

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(ii) Choose **one** of the variables you have identified in (i).

State the effect it could have on the results if the measured values were not similar.

(1)

Variable

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The effect it could have on the results if the values were not similar.

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(Total for Question 2 = 10 marks)
